

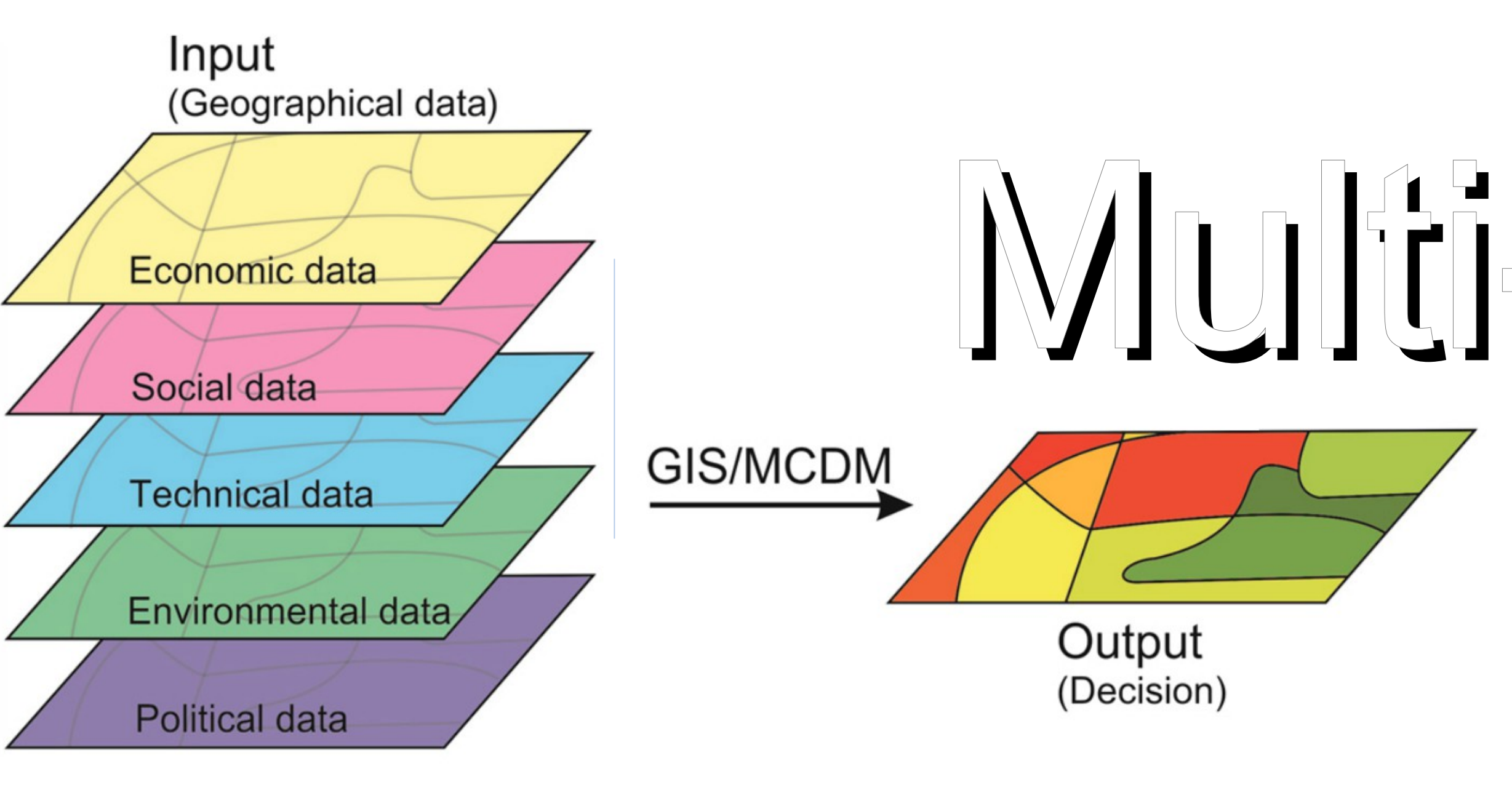


Figure 1: a conceptual model of MCE in GIS (Rikalovic, and Cocić, 2014)



Figure 2: potentially conflicting landuses (1)

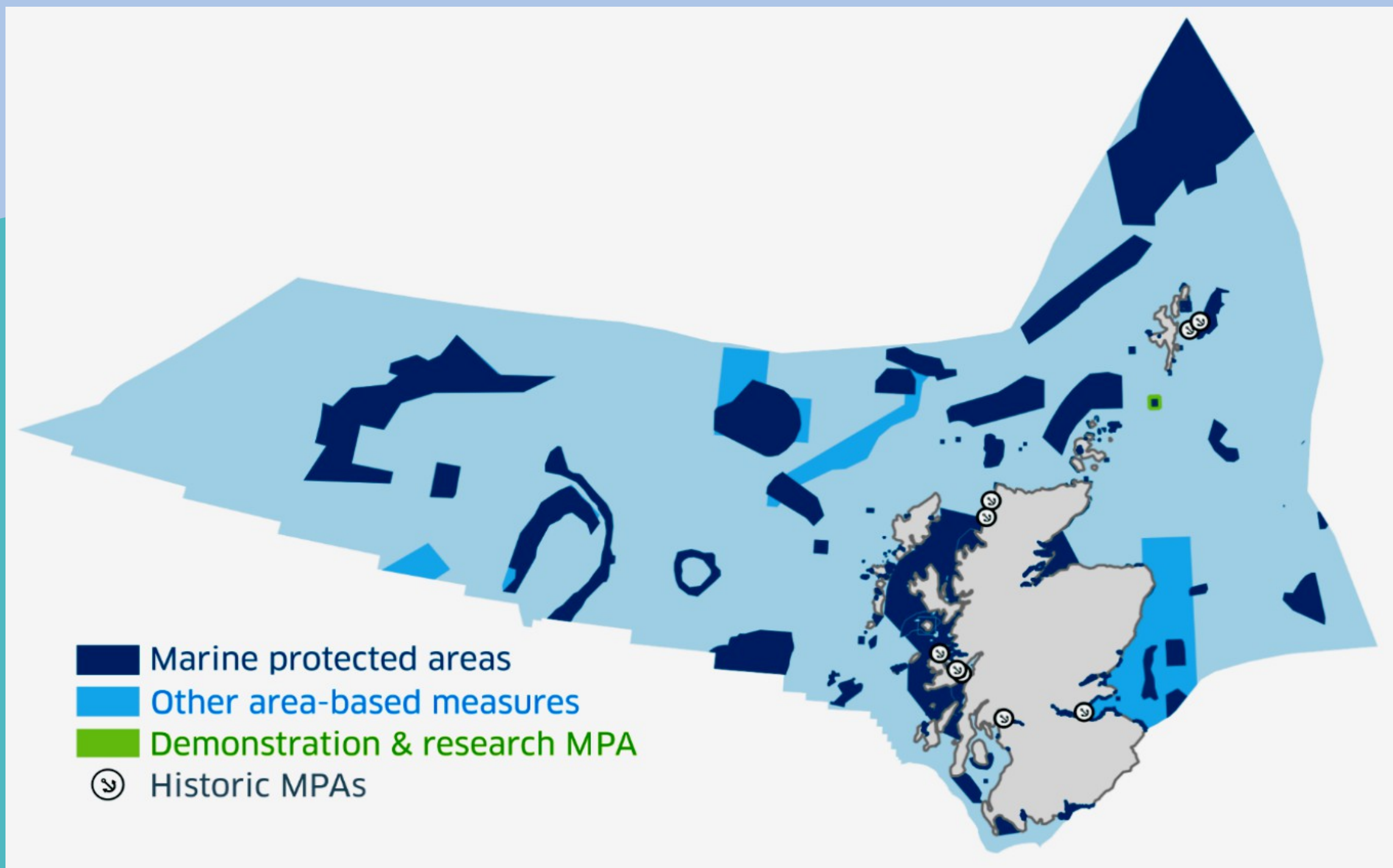


Figure 3: an example of sites selected for preservation: Marine Protected Areas (Scottish Government, 2018)

4. Ordered weighting and decision strategy space

Multi-criteria decisions involve trade-off and risk: criteria can offset one another, and highly suitable locations can still have low criteria scores. Risk and trade-off in MCE form the 'decision strategy space'. Their prevalence can be modified with Ordered Weighting: assigning a second set of weights to criteria ranked to value for each unique location. Assigning low rank criteria a high weight reduces risk, they will lower the suitability of the location; the opposite is true for high rank criteria. Trade-off is influenced by the difference in ordered weight values, a narrow focus avoids trade-off and an even spread does the opposite. Figure 10 shows the effect of modifying risk and trade-off in an MCE: higher risk allows for more locations to become suitable, higher trade-off allows for more flexibility in assigning suitability status.

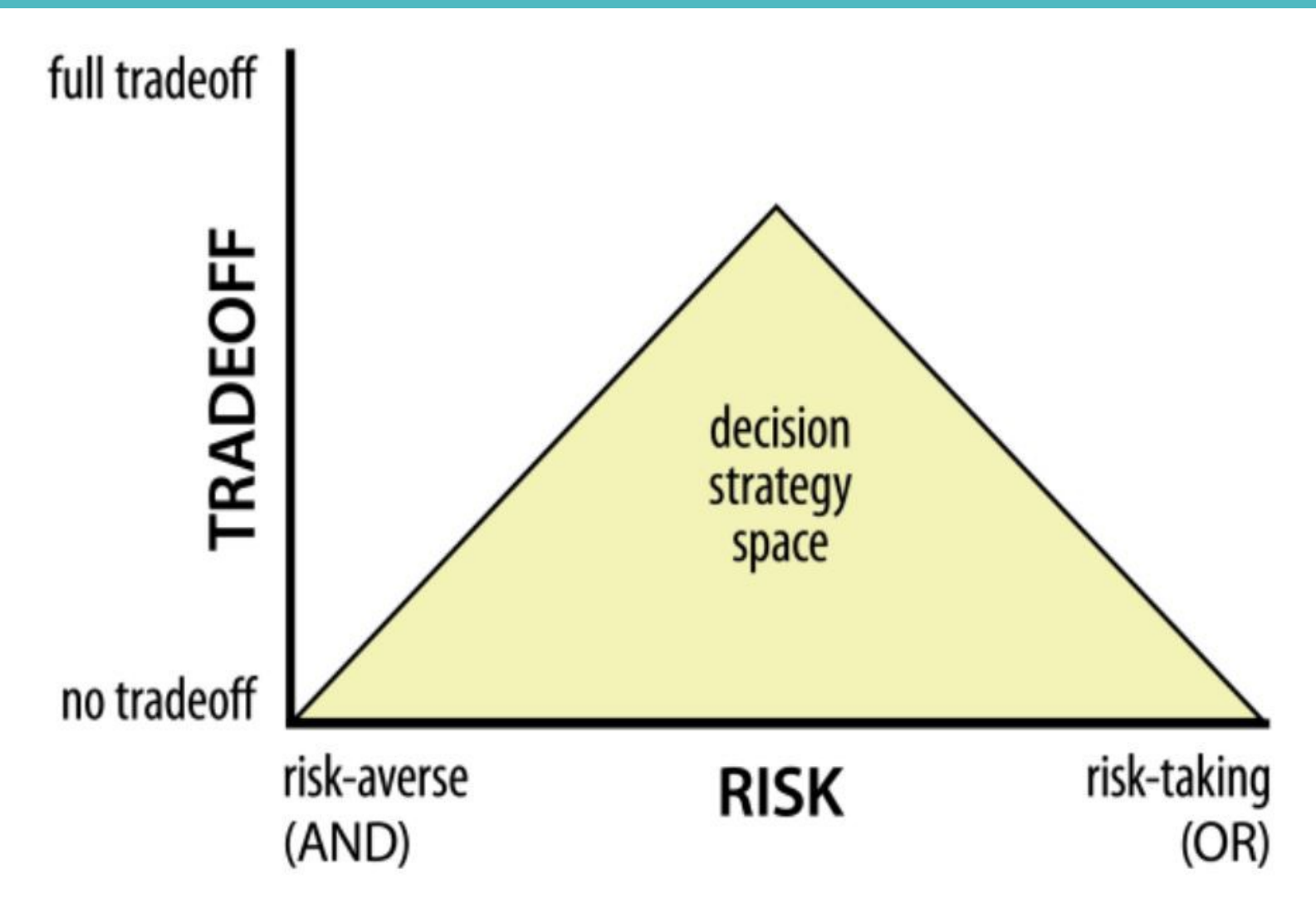


Figure 9: Decision Strategy Space. The X-axis shows which ranked criteria are given more focus, from low (left) to high (right). The Y-axis shows the focus of increased weights, from single values (below) to spread evenly (above) (Eastman, 2015)

2. Constraints and criteria

That which makes a location more or less suitable is to be examined, clarified and possibly quantified. Criteria with spatial variability can be mapped, in Boolean (yes/no) or continuous maps. Examples of both are shown in figure 4. Their variation will be determined by the occurrence or avoidance of conditions (e.g. solid subsoil), closeness to an element (e.g. to waterways), or a relation to an element through an intermediate criterion like cost distance (determined by the road network and its quality). Even within the different spatial distributions there are many variations. They can be visualised as plots relating the suitability of a site and the value of a criterion (e.g. distance to a feature). Figure 5 shows some examples of linear (1&2), J-shaped (3&4), and sigmoid (5&6) relationships.

Figure 4: four maps showing criteria spatial variation. Buffers are an example of a Boolean distribution of yes/no per location. The other maps show continuous variation.: the Distance map could equate to curve 1 or 3 from figure 5, and the cost distance does not have a single curve but is dependant on other spatial criteria, in this case roads and their quality (Eastman, 2015)

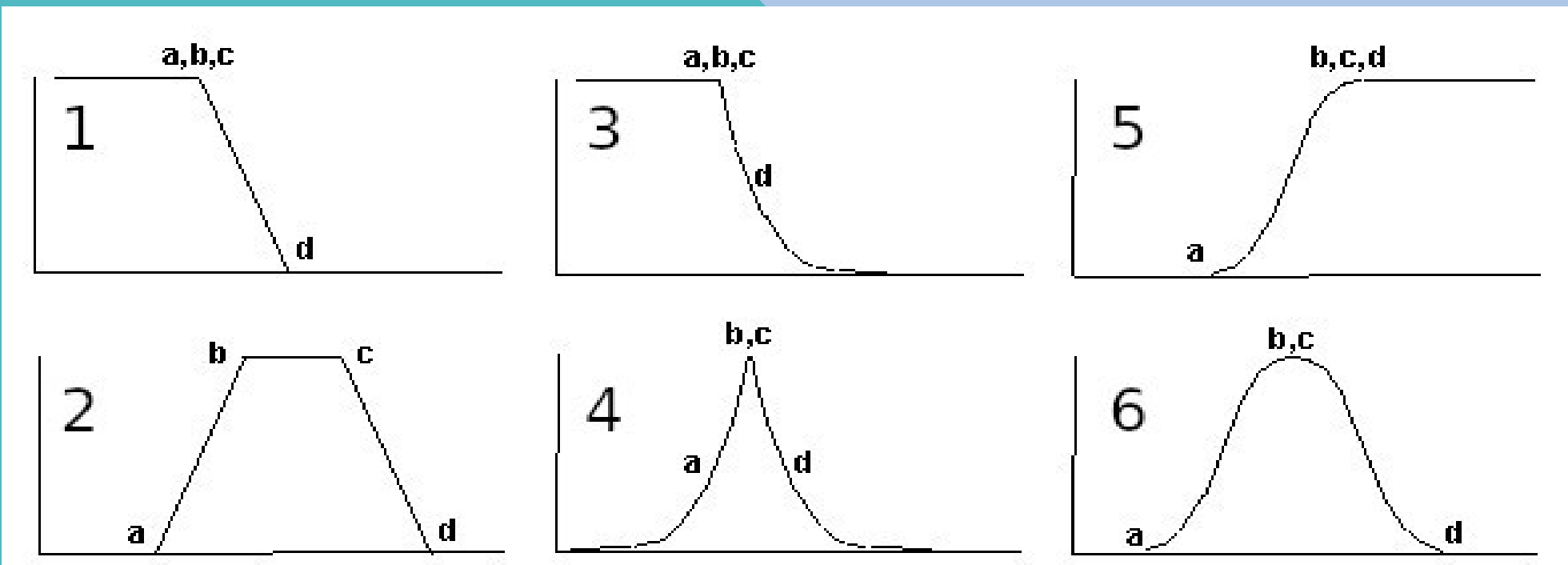
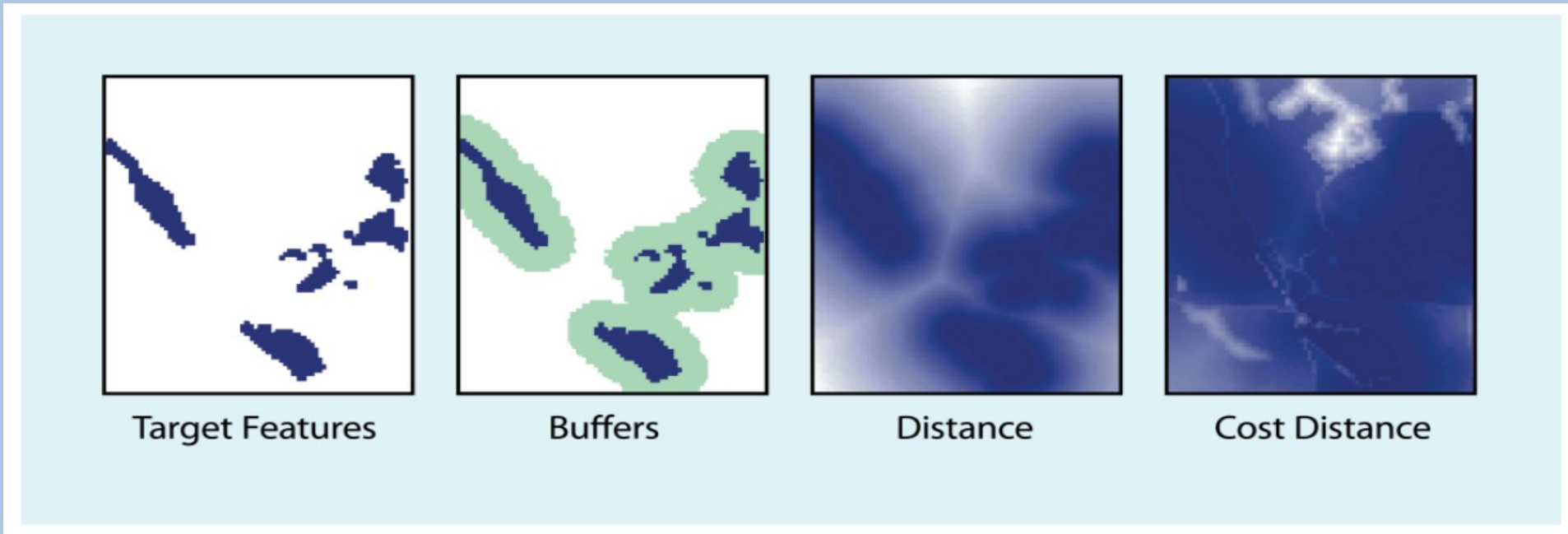


Figure 5: examples of spatial variation of criteria. The Y-axis represents suitability, the X-axis criterion value. The figures show suitability on the vertical axis from 0 to 1, and the quantity of the criterion on the horizontal axis. A, b, c, and d represent points at which the suitability rises from 0, reaches 1, drops to and reaches zero respectively, if applicable. (online source 'Fuzzy')

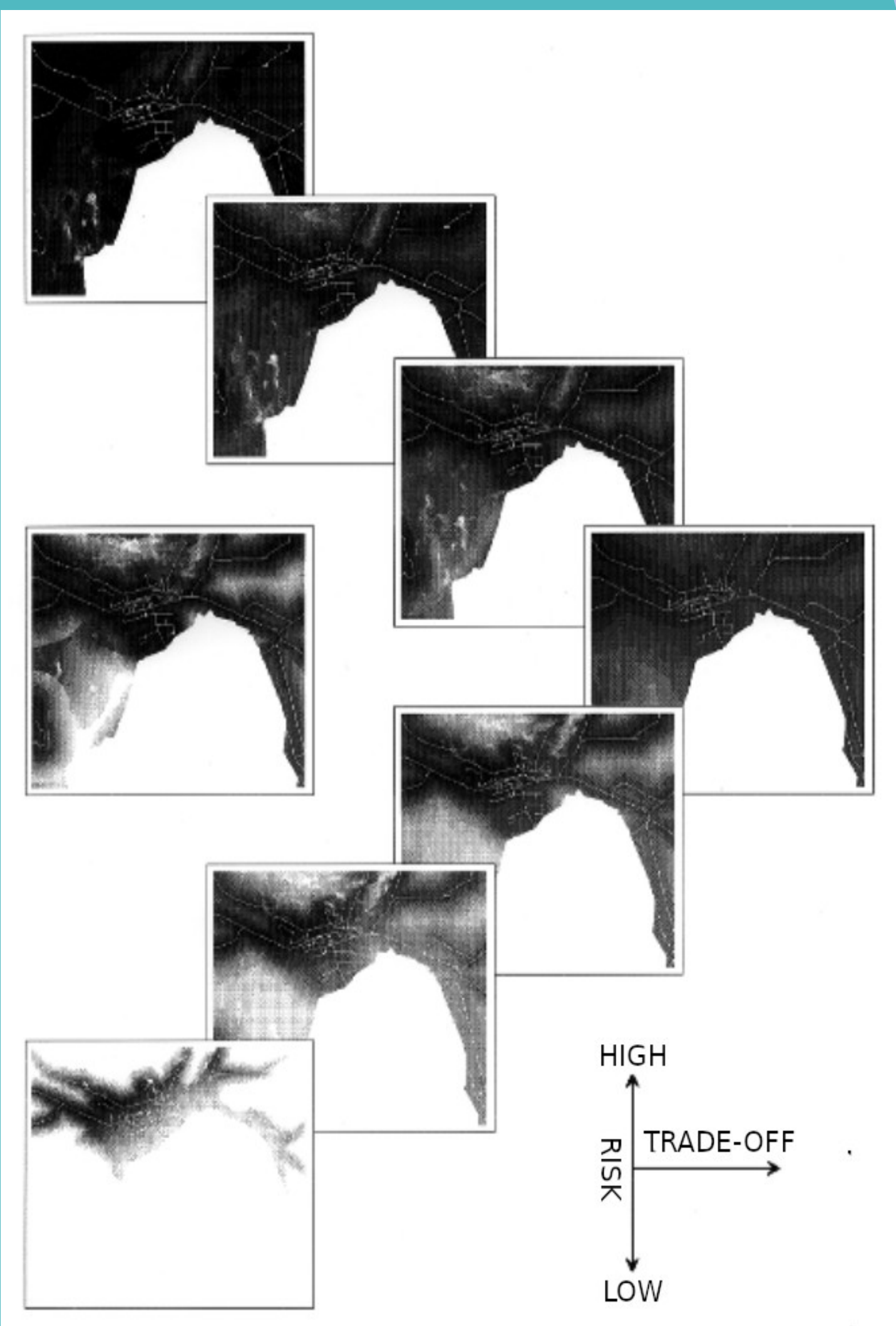


Figure 10: the effect of modifications of risk and trade-off in an MCE. (Eastman, 1999)

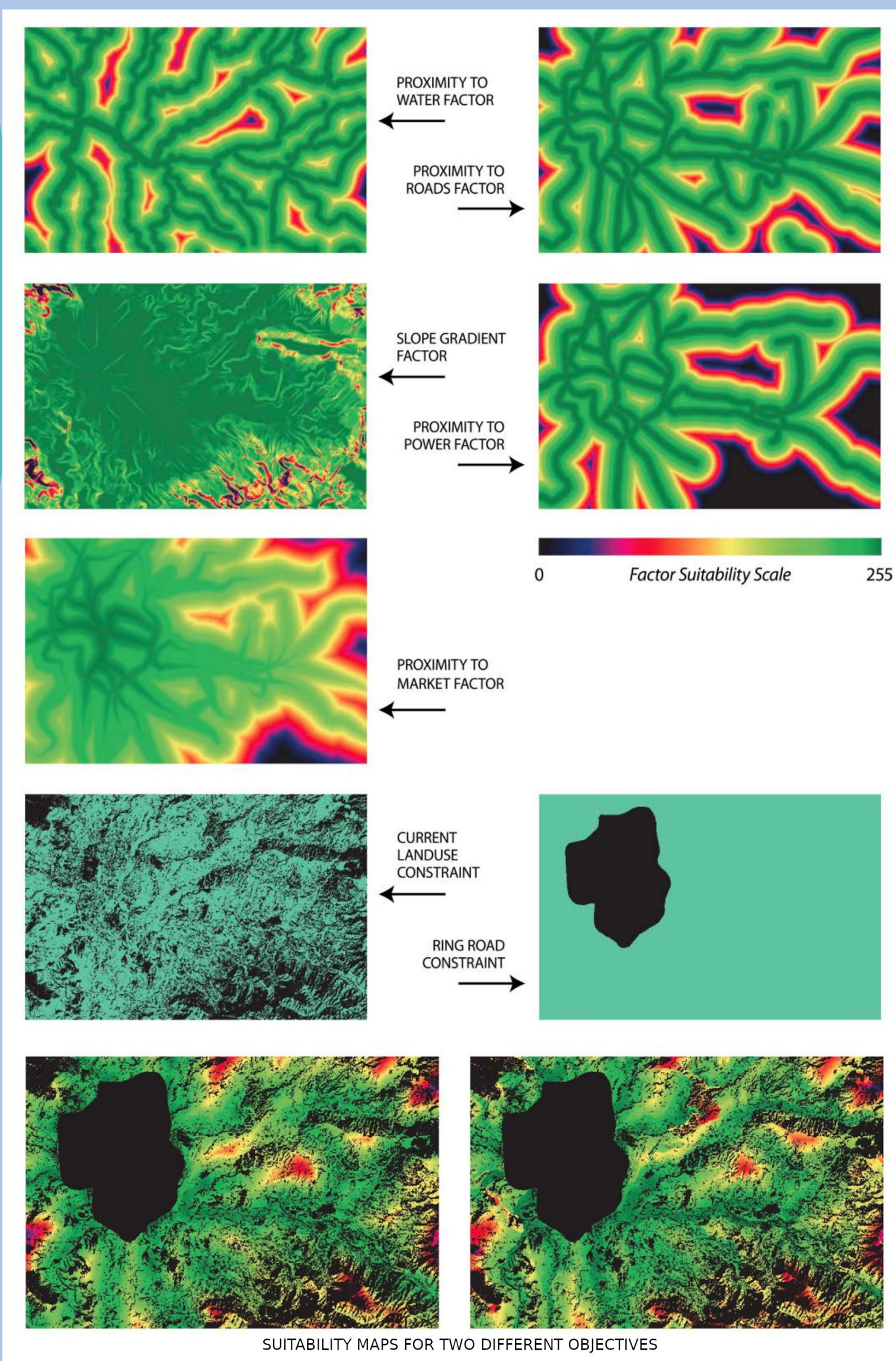


Figure 11: a full example of the MCE process (Eastman, 2015)

3. Weighting



Factor	Derived weight
Proximity to roads	0.3770
Proximity to labour force	0.0979
Slope gradient	0.3605
Distance from wildlife reserves	0.1647

Figures 6, 7 and 8: a pairwise comparison matrix, derived weights (Eastman, 1999; derived from Saaty, 1977) and a boardroom (2)

To reflect relative priorities in criteria, weights are assigned to each. A structured method for this has been proposed by Saaty (1977): the pairwise comparison matrix. An example is shown in figure 6; several typical criteria have been entered. Each cell represents the relative importance of the criterion in the row over that in the column. Completing this matrix allows for comparison of each pair of criteria separately, and subsequent integration in one group by calculating the matrix' principal eigenvector and determining individual weights from it. The weights for this example are shown in figure 7. To result in a singular weighted suitability score, the separate weights always sum to 1. Note that weighting will apply to the entire range of a criterion, so the points where suitability starts to rise above zero must be chosen to also be represented by the assigned priority values. Criteria cant always be compared; they might originate with stakeholders with conflicting interests, or they might pertain to different interest topics such as environmental values and industrialisation priorities. These criteria can be weighted to each other in groups, after which a decision on weight for the groups can be made. This is the stage in the process where uncertainty, conflict of interest, or inconsistencies can be clarified, and the internal logical coherence of the final decision is generated. Much of this stage of the analysis will consist of discussion.

5. Suitability mapping

The weighted criteria can now be integrated into a single map that shows suitability for each location. Multiple maps can be generated for different sets of weights, or for different weight grouping or ordering. An example of the MCE process from multiple criteria and constraints to two suitability maps can be seen in figure 9 to the left.

6. Site selection

From the suitability map, selections can be made that best fulfil set requirements such as a minimum size, interconnectedness, or spread. The selection characteristics might depend on the amount of suitable locations found. Several maps might be generated with slightly different weights to examine the variation in suitability. The values of the criteria at the chosen sites may be extracted and studied to understand how the MCE arrived at these conclusions.

Sources

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